

TypeScript monorepo: MCP server + stock screener API + shared trading/core utilities

What it is

A set of developer-focused trading services: an MCP (Model Context Protocol) server for AI-assisted workflows and a REST stock-screener API.

Core integrations use Alpaca for market data, portfolio, and order operations, with optional data providers (Yahoo, Polygon, Finnhub, FRED, Finviz).

Who it's for

Developers building AI-assisted trading tools (MCP clients) and teams needing a small HTTP API for quotes and screening.

How to run (minimal)

- Prereqs: Node.js >= 20 and pnpm.
- Install/build: `pnpm install` then `pnpm build`.
- Set env: `ALPACA_API_KEY`, `ALPACA_API_SECRET` (optional: `ALPACA_PAPER=true` for paper trading).
- Run MCP server (stdio): `pnpm dev` (or `pnpm --filter @trading/mcp dev`).
- Run screener API: `pnpm dev:screener` (defaults to `PORT=3000`).
- MCP client configuration snippet (e.g., Claude Desktop):
Not found in repo.

What it does

- MCP tools for quotes, OHLCV bars, and market status (Alpaca-backed).
- Portfolio/account tools and order management (place/get/list/cancel orders).
- Technical indicators (RSI, MACD, SMA/EMA, Bollinger Bands) and basic signal generation.
- Stock screening helpers (movers, RSI extremes, unusual volume, highs/lows, trending, symbol search).
- REST API for quotes/batch quotes/bars, plus scan criteria, movers, and signals endpoints.
- Unified provider layer with fallbacks and `/health` checks for configured data sources.

How it works (repo-based)

- `@trading/core` (`packages/core`): config/env, Alpaca client, provider adapters, caching and rate limiting.
- `@trading/mcp` (`packages/mcp`): MCP server over stdio; tool handlers in `src/tools/*` call `@trading/core` and return JSON.
- `@trading/screener` (`packages/screener`): Hono app (`src/app.ts`) with quote + screener routes; `ScreenerService` pulls snapshots/bars and computes RSI/SMA.
- Flow (MCP): MCP client -> stdio transport -> tools -> core -> external APIs.
- Flow (HTTP): client -> Hono routes/service -> core -> external APIs.